

Homework 2

Jizhou Yan ^{TA}

Instructions before the assignment 作业前须知:

- This assignment is for the **third week's homework** of the System Dynamics and Control course (hereafter referred to as SDC). The submission deadline is **before 11:59 PM on October 10, 2024**. Please ensure timely submission, as late submissions will result in point deductions. 本次作业为 SDC 第三周课程作业, 提交时间为 **2024/10/10 23:59** 前。
 - If submitted within 12 hours after the deadline, no points will be deducted (**this is allowed only twice, so focus on completing it properly rather than rushing to meet the deadline if you're unable to finish on time**);
 - If submitted within three days after the deadline, **20% of the total score of this assignment will be deducted**;
 - If submitted within two week after the deadline, **30% of the total score of this assignment will be deducted**;
 - Any other late submissions will result in a **40% deduction**;
- For late submissions, please contact the teaching assistant(TA), **Yan Jizhou**, via email at **yanjizhou312@163.com** or WeChat **y3139838452**. Make sure to use the filename format as "SDC-Assignment-x-Resubmission-Name-Student ID." 作业补交时通过邮箱或微信与助教闫济洲联系, 邮件标题及文件名备注为 "SDC-第 x 次作业补交-姓名-学号".
- The assignment is graded out of 10 points. It consists of basic course questions (10 points) and extension questions (0 to 5 points). **The final assignment score for each submission will be the smaller value between 10 and the actual score.** Each assignment earns all points by completing only the basic assignment. 每次作业仅完成基础作业即可获得所有分数, 鼓励同学们完成拓展课程题目。
- Students are encouraged to attempt the extension questions. Any extra points earned will be proportionally added to the final course grade for possible grade improvement.
- **Academic misconduct such as plagiarism is strictly prohibited in course work. Plagiarism and other academic misconduct will be dealt with in accordance with the relevant regulations of the University.** If you have discussed the assignment with classmates or received help, please note at the beginning of the assignment: "This assignment was completed after discussion with XXX (classmate's name), and part of the thought process was referenced from XXX (classmate's name)." **Every contribution deserves recognition. Unacknowledged similarities between assignments will be treated as plagiarism.** 课程作业中严禁抄袭等学术不端行为。如作业完成过程中经历了同学讨论/同学帮助的情况, 请在作业最前面注明 "本次作业经过与 XXX (同学姓名) 讨论, 参考了 XXX (同学姓名) 的部分思路"。每位同学的帮助都值得被记录。不被注明的作业雷同将被认为是抄袭行为。

1. Problem 1

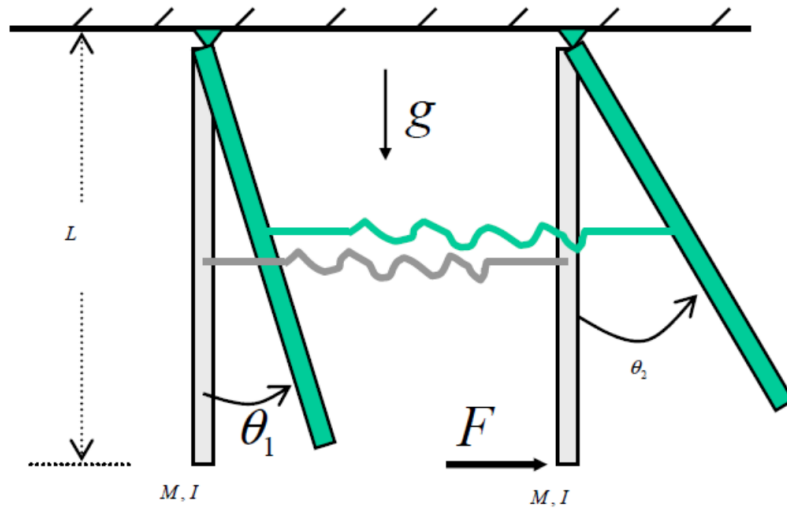
For these LTI systems with following characteristic polynomial, find the range of K to stabilize these systems.

1. $D(s) = s^4 + 22s^3 + 10s^2 + 2s + K$
2. $D(s) = s^4 + 20Ks^3 + 5s^2 + (10 + K)s + 15$
3. $D(s) = s^3 + (K + 0.5)s^2 + 4Ks + 50$
4. $D(s) = s^4 + Ks^3 + s^2 + s + 1$

2. Problem 2

Consider the double pendulum system studied in Homework2. For this problem, you can use the Mathematical tool software like Matlab or Mathematica.

Assuming that $M = 1, g = 10, L = 1, I = ML^2/3, K = 1$ and $B_s = 0.01$:



1. Obtain the transfer function of the system from the input F to the output Θ_1 . Obtain the poles and zeros of the transfer function. (For the dynamics equations, refer to the tips below)
2. Obtain the free response of the system for $\theta_1 = 0.1, \dot{\theta}_1 = 0, \ddot{\theta}_1 = 0$, and $\ddot{\theta}_1 = 0$. Plot the obtained free response for a time duration of 100 sec.
3. Obtain the forced response of the system for the unit impulse input $F = \delta(t)$ with zero I.C. Plot the obtained forced response.
4. Obtain the forced response of the system for a sinusoidal input with unit amplitude and an input frequency of 3rad/sec (i.e., $u(t) = \sin(3t)$) with zero I.C. Obtain the steady-state response of the system (i.e., $y_{ss}(t) = \lim_{t \rightarrow \infty} y(t)$) as well. Plot the obtained forced response.
5. Obtain the stability of the system.